

Assignment 3: Diffusion Playground

Sampling, Editing, and Illusions with a Pretrained Text-to-Image Diffusion Model (DeepFloyd IF)

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Part 0: Loading the Model

Task 0.1: Sampling with your own prompts + inference-step comparison

Prompts used (all exact keys in the precomputed embedding dictionary):

- "an oil painting of an old man"
- "an oil painting of people around a campfire"
- "a rocket ship"

Random seed: 100 (set with `seed_everything(100)`, and reused for all later parts).

Stage 1 (64×64):



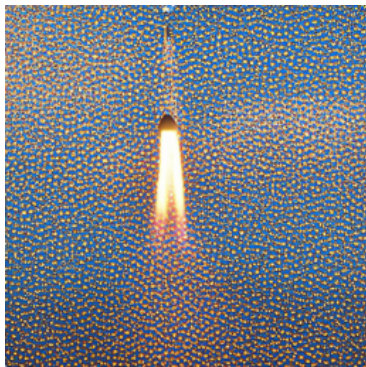
old man / campfire / rocket ship — stage 1

Stage 2 (upsampled to 256×256):



old man / campfire / rocket ship — stage 2

Low vs. high num_inference_steps (prompt "a rocket ship"):



5 steps (left) vs. 50 steps (right)

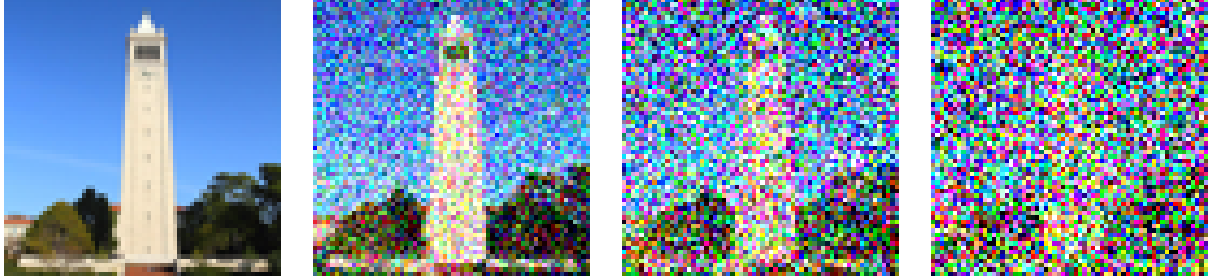
Discussion. All three outputs match their prompts. The old man is a bearded, wrinkled bust with visible brushwork; the campfire prompt gives five figures seated around a flame in a pine forest; and the rocket is an unmistakable finned rocket climbing on a plume. The “oil painting” wording clearly registers in the first two: both come out with a painted, textured look rather than a photographic one.

- *Stage 1 vs. Stage 2.* Stage 1 fixes the composition at 64×64 — the content is already correct but soft. Stage 2 is a conditional super-resolution diffusion model, and at 256×256 it fills in the edges and texture (the old man’s beard hairs, the flame detail) without changing what the image depicts.
- *5 vs. 50 steps.* The difference is stark on the rocket. At 5 steps the sampler can’t clear the noise in time: the result is mostly blue/orange speckle with only a faint vertical flame streak showing through, barely readable as a rocket. At 50 steps each reverse step is a smaller, more accurate update and the rocket comes out clean and saturated. (Both stages ran at the same step count here, so this combines stage-1 and stage-2 quality.)

Part 1: Sampling-Loop Mechanics

Task 1.1: The Forward Process

The forward process adds noise per the pretrained schedule: $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$. Test image noised at $t \in \{250, 500, 750\}$:



original / $t = 250$ / $t = 500$ / $t = 750$

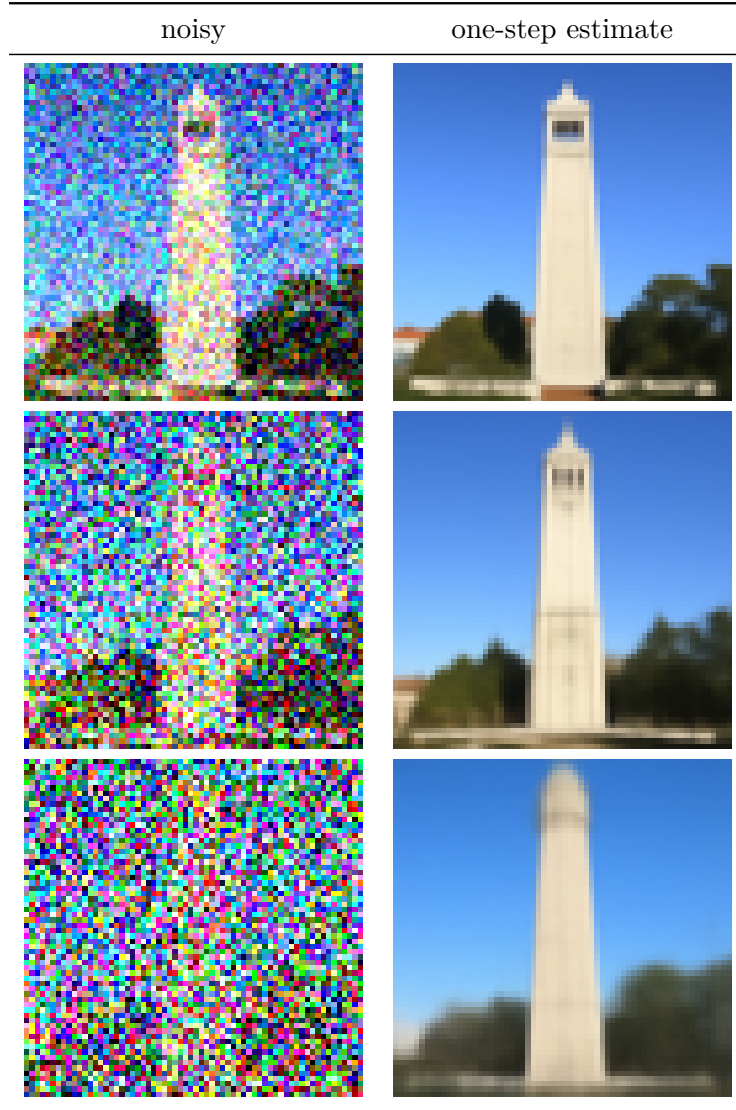
Discussion. As t grows, $\sqrt{\alpha_t}$ shrinks and $\sqrt{1 - \alpha_t}$ grows, so the clean signal is progressively drowned out by Gaussian noise. At $t = 250$ the structure is still clearly visible; by $t = 750$ the image is almost pure noise.

Task 1.2a: One-Step Denoising

Using the UNet's noise estimate to jump straight to a clean-image estimate: $\hat{x}_0 = (x_t - \sqrt{1 - \alpha_t} \hat{\varepsilon}_\theta(x_t, t)) / \sqrt{\alpha_t}$. Null prompt "a high quality photo", at $t \in \{250, 500, 750\}$:



original



$t = 250$ (top), $t = 500$ (middle), $t = 750$ (bottom)

Discussion. At $t = 250$ the one-step estimate is already a clean campanile. As t rises it falls apart: at $t = 500$ the tower is soft and washed out, and at $t = 750$ it is a blurry, low-contrast blob. This is expected, since the model has to reconstruct the whole clean image from almost pure noise in a single shot, and at high noise there is not enough signal left to do that.

Task 1.2b: Iterative Denoising (strided schedule)

`strided_timesteps = list(range(990, -1, -30))` → **34 timesteps** (990, 960, ..., 30, 0), following the DDPM reverse step

$$x_{t'} = \frac{\sqrt{\bar{\alpha}_{t'}} \beta_t}{1 - \bar{\alpha}_t} \hat{x}_0 + \frac{\sqrt{\bar{\alpha}_t} (1 - \bar{\alpha}_{t'})}{1 - \bar{\alpha}_t} x_t + \sigma_t z,$$

with $\alpha_t = \bar{\alpha}_t / \bar{\alpha}_{t'}$, $\beta_t = 1 - \alpha_t$. The iterative loop was run starting from a test image noised to `i_start = 10`. (The noisy `i_start = 10` image and the iteratively denoised result are shown side by side in Task 1.2c below.)

Task 1.2c: One-Step vs. Iterative (same noise level, `i_start = 10`)



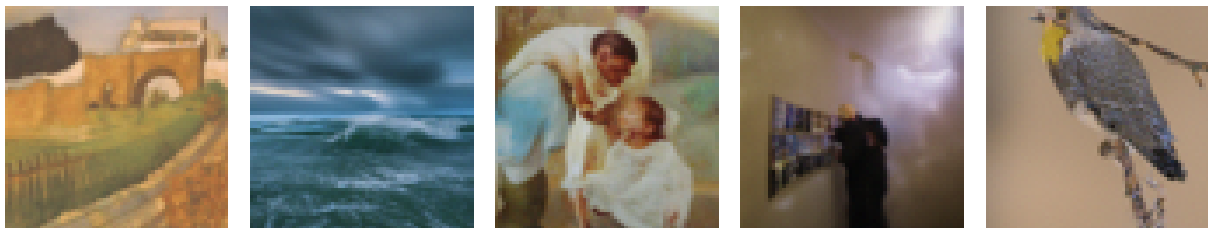
noisy (`i_start=10`) / iterative / one-step

Discussion: why iterative denoising beats one-step with the same model. The one-step panel on the right is a smeared, ghostly campanile, while the iterative one in the middle is a crisp tower against a clear sky. At this noise level many different clean images could have produced the same noisy x_t , and the UNet only outputs a single noise estimate, so its one-shot answer lands near the average of all of them — which is why it looks washed out. The iterative version removes a little noise per step; after each step the image is slightly cleaner, fewer clean images are still consistent with it, and the model can commit to one and start filling in real detail. The weights are identical in both runs, so the gain comes entirely from splitting the work into many small steps rather than one big jump.

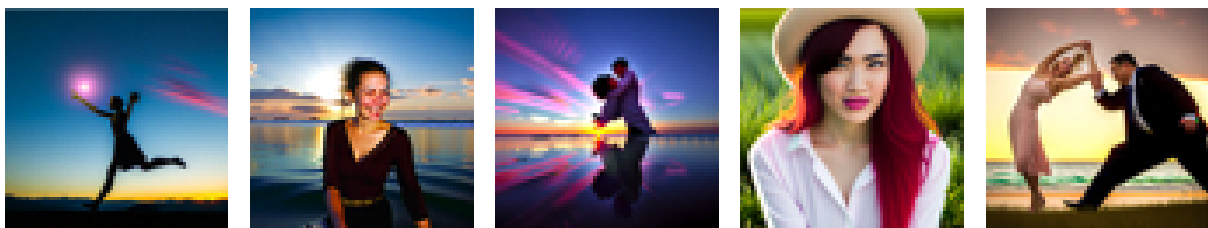
Task 1.3: Sampling With and Without CFG

Prompt "a high quality photo"; CFG uses $\varepsilon_{\text{cfg}} = \varepsilon_u + \gamma(\varepsilon_c - \varepsilon_u)$ with $\gamma = 7$.

Without CFG (5 samples):



With CFG, $\gamma = 7$ (5 samples):



Discussion. The no-CFG row is weak: the samples are dull and low-contrast, and a couple barely read as photos. The $\gamma = 7$ row is a clear improvement: sharper, more saturated, and more convincingly photographic, because guidance pushes away from the unconditional estimate

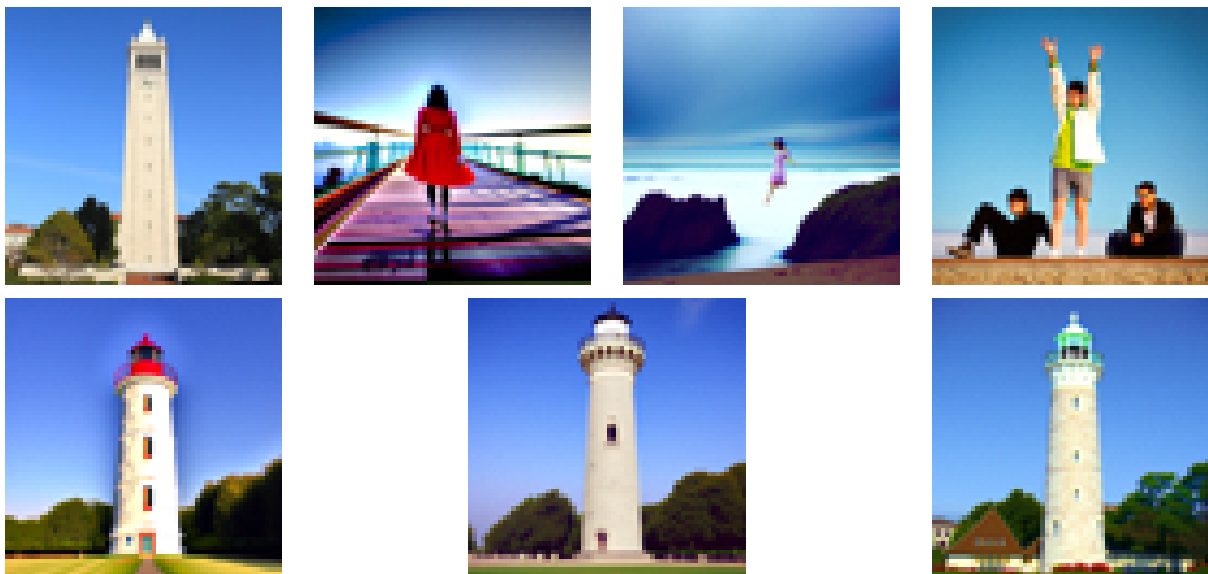
and leans harder into the conditional one. The cost is diversity: the five CFG samples resemble each other more than the five un-guided ones do.

Part 2: Image Editing

Task 2.1: SDEdit on the provided image and on your own image

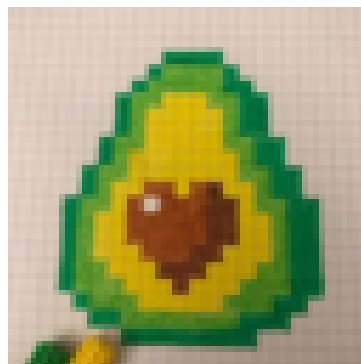
Procedure: noise the input to the timestep for each $i_start \in \{1, 3, 5, 7, 10, 20\}$, then run `iterative_denoise_cfg` from that point with the null prompt "a high quality photo".

Provided test image (campanile):

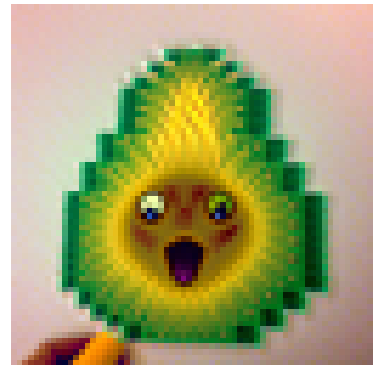
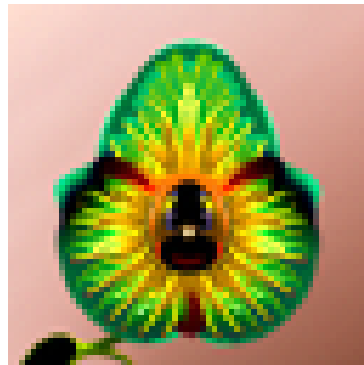
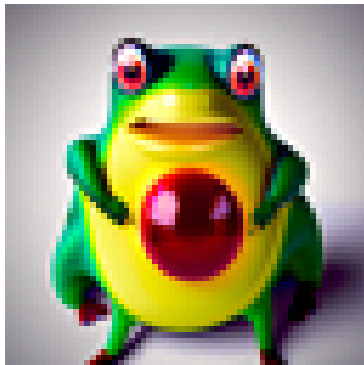


original / $i_start = 1, 3, 5, 7, 10, 20$

Own image, web image (source: a downloaded web image):

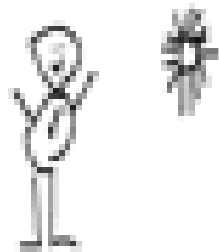


web source

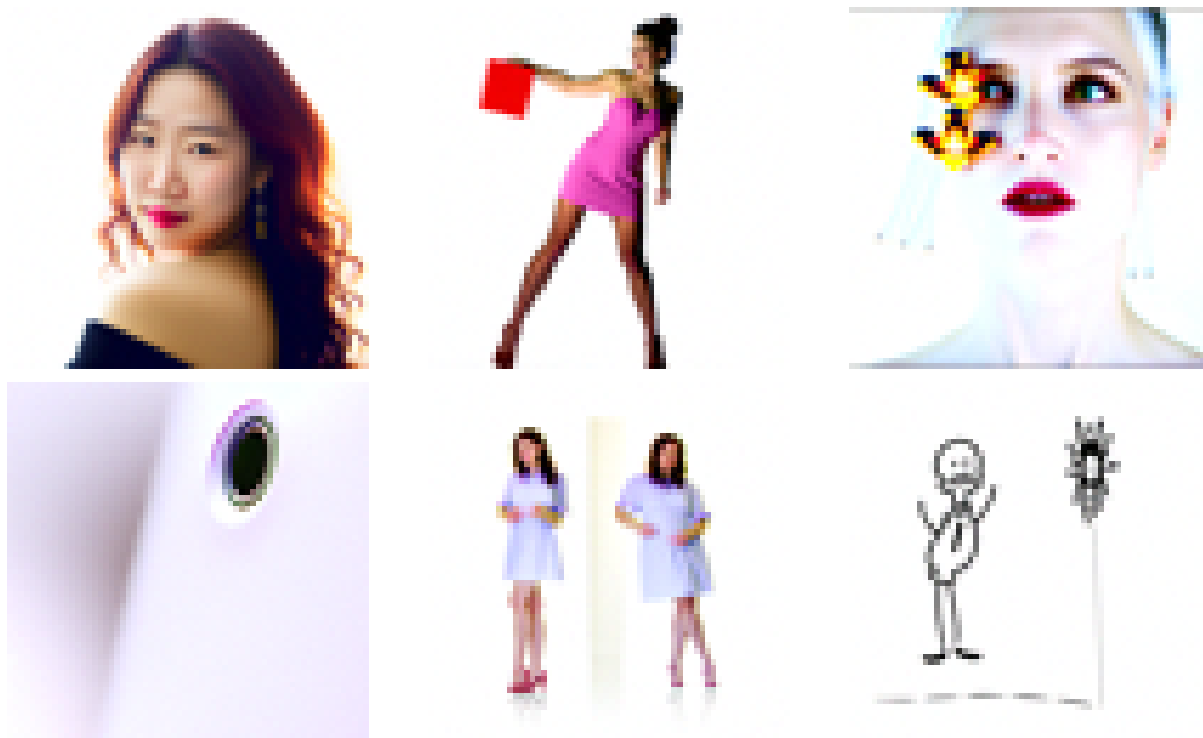


$i_start = 1, 3, 5, 7, 10, 20$

Own image, hand-drawn:



hand-drawn source



$i_start = 1, 3, 5, 7, 10, 20$

Discussion: edit strength vs. i_start . i_start controls how much noise we add before denoising again, i.e. how far back in the schedule we jump in. Because `strided_timesteps` runs from 990 down to 0, a *small* i_start corresponds to a *high* timestep: at $i_start=1$ we jump in at $t \approx 960$, which destroys almost all of the input, so the model has to reinvent large chunks and produces a strong, creative edit that wanders well away from what we fed it. As i_start grows the starting timestep drops ($i_start=20$ is $t \approx 390$), less noise is added, and more of the original signal survives, so by $i_start=20$ the output stays close to the input. On the campanile this shows up as a nice progression: at $i_start=7$ the tower comes back with a red-and-white lighthouse top, at $i_start=10$ it is a slimmer white lighthouse, and by $i_start=20$ it has snapped back to essentially the original campanile. In short, *small* i_start gives the strongest, most creative edit and *large* i_start keeps you faithful to the original. This is clearest on the hand-drawn input, which only turns into a realistic photo at the *small*- i_start settings and stays close to the drawing at $i_start=20$.

Task 2.2: Inpainting

New content is generated only inside a binary mask (1 = regenerate); everything outside is forced back to the original image, re-noised to the current timestep, at every step.

Provided test image (mask over a region):

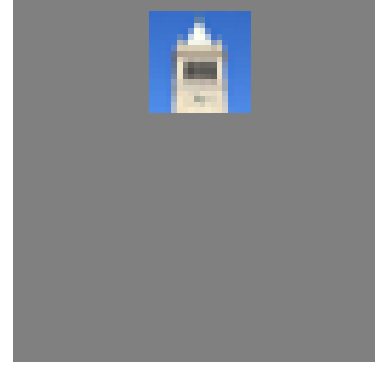
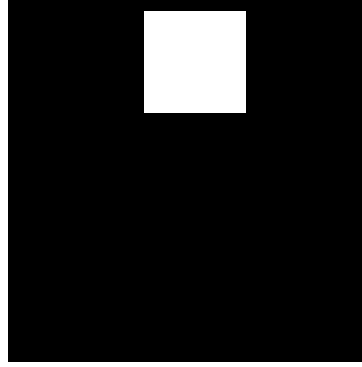
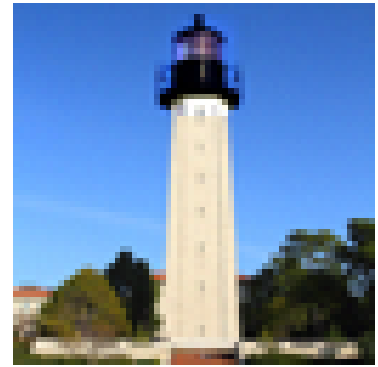
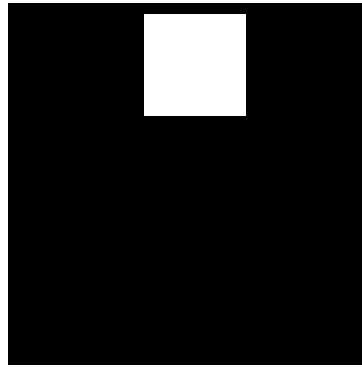
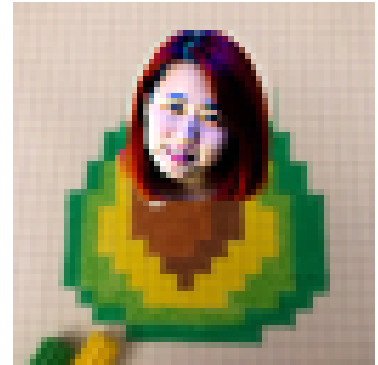
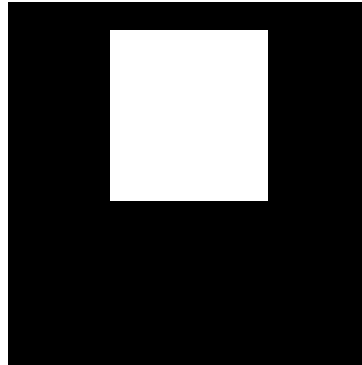
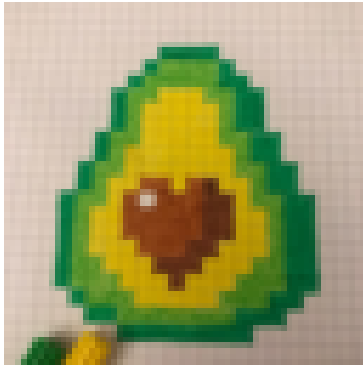


image / mask / to replace



original / mask / inpainted

Own image, own mask (inpainting the web image with a custom rectangular mask):



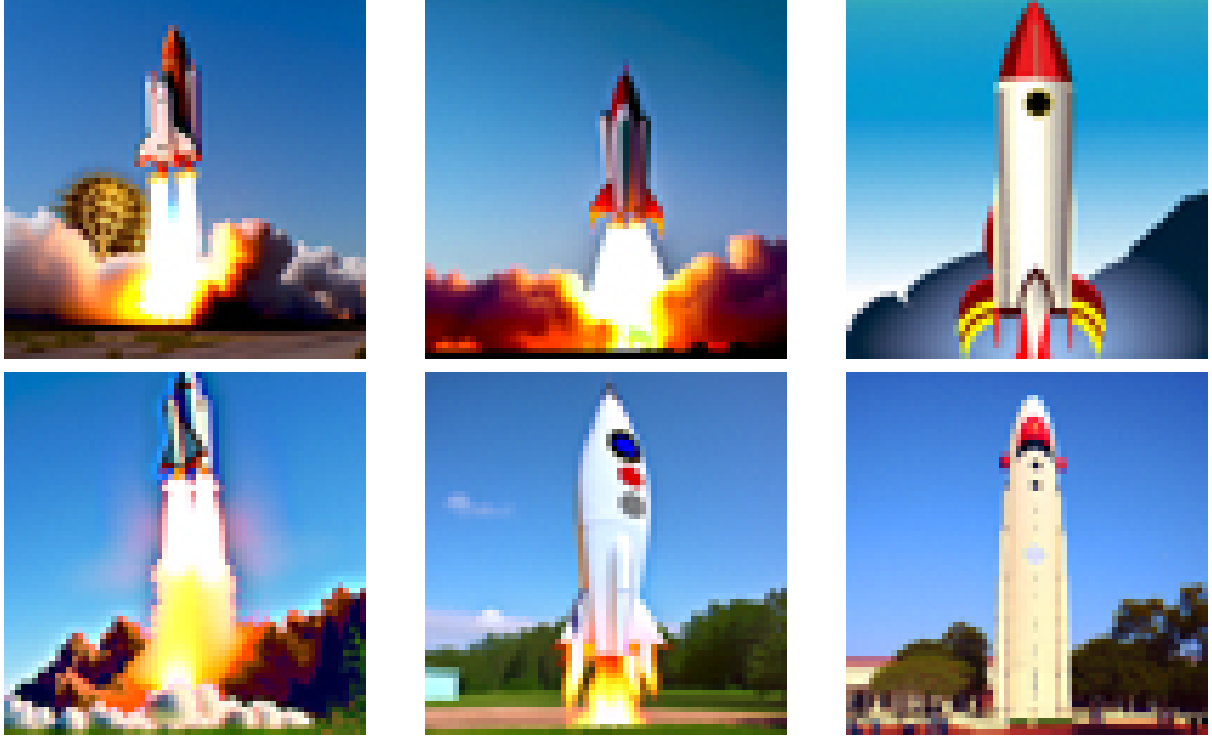
original / mask / inpainted

Discussion. The reverse process runs normally inside the mask, but at each step the region outside the mask is overwritten with the original image noised to match the current timestep. This keeps the surroundings fixed while the model generates new content inside the mask. On the campanile the masked top is regenerated as a lighthouse-style lantern room, and it blends in reasonably: the surrounding sky and tower are untouched, though a faint seam is visible along the *bottom* edge of the mask (the left and right edges run through flat sky and are effectively invisible). The custom mask on the avocado is a good illustration of the method's limits: with only the generic "a high quality photo" prompt, the model fills the masked pit region with a realistic woman's face, which is locally coherent but clashes with the flat pixel-art style

around it. So the outside stays fixed by construction, but what gets generated inside is only as context-appropriate as the prompt allows.

Task 2.3: Text-Conditioned Image-to-Image Translation

Same as SDEdit but steered by a real prompt instead of the null prompt. **Prompt used: "a rocket ship"**, $i_start \in \{1, 3, 5, 7, 10, 20\}$ on the provided test image.



$i_start = 1, 3, 5, 7, 10, 20$

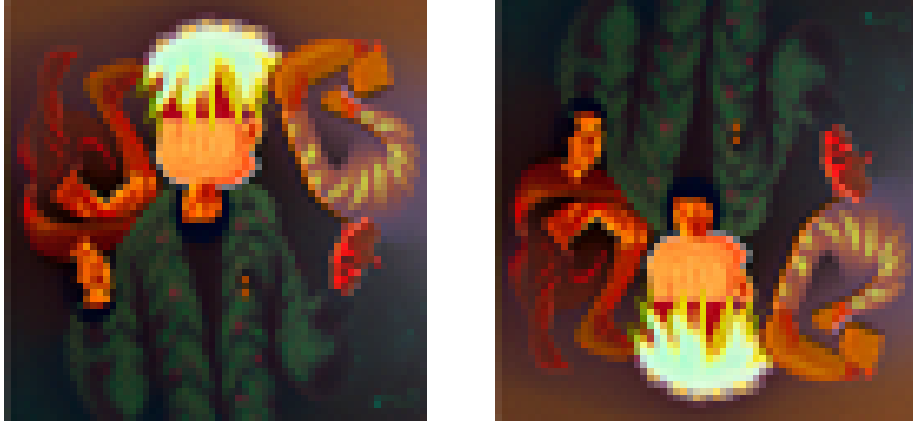
Discussion. Compared with the null-prompt edit in Task 2.1, the text prompt injects *semantic* content, not just “realism.” At $i_start=1$ the input is almost fully noised, so the model is free to build a complete rocket-launch scene: a finned rocket on a colored plume against a clear blue daytime sky, with a bright yellow burst at upper left. As i_start *increases* more of the campanile survives and pulls the result back toward the tower; by $i_start=20$ the campanile has returned, but with a red-and-white rocket nose cone on top — something the null-prompt run at the same i_start in Task 2.1 never produces, so the prompt is still biting even when the input strongly constrains the output. So the prompt decides *what* the edit turns into, while i_start decides *how strongly* the original image constrains it.

Part 3: Diffusion Illusions

Task 3.1: Visual Anagrams

Combined noise estimate $\varepsilon = \frac{1}{2}(\varepsilon_\theta(x_t, t, p_1) + \text{flip}(\varepsilon_\theta(\text{flip}(x_t), t, p_2)))$, $\gamma = 7$, denoised from pure noise ($i_start = 0$).

- p_1 (upright) = "an oil painting of an old man"
- p_2 (flipped) = "an oil painting of people around a campfire"



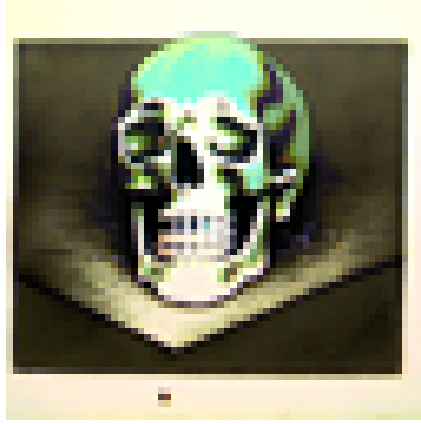
upright: an old man / flipped: people around a campfire

Discussion. Averaging the two noise estimates (one on the image, one on its vertical flip) pulls the reverse process toward both prompts under their respective orientations. In our result the effect is uneven: the *flipped* orientation reads clearly as people around a campfire — a bright flame low-center with orange figures around it — while the *upright* orientation is the weaker of the two, coming out as loose oil-painting forms with a pale blob near the top that only loosely suggests a head rather than a convincing old man. So this sample is only a partially successful visual anagram: the campfire orientation is clearly recognizable, but the upright old-man orientation is weak. A cleaner result would likely need trying a few more seeds and keeping the best.

Task 3.2: Hybrid Images (Bonus)

Combined estimate $\varepsilon = f_{\text{low}}(\varepsilon_{\theta}(x_t, t, p_{\text{low}})) + f_{\text{high}}(\varepsilon_{\theta}(x_t, t, p_{\text{high}}))$, where f_{low} is a Gaussian blur (kernel_size=9, sigma=2.0) and f_{high} is the estimate minus its own low-pass. $\gamma = 7$, i_start = 0.

- p_{low} (low frequencies / far) = "a lithograph of waterfalls"
- p_{high} (high frequencies / near) = "a lithograph of a skull"



hybrid (far: waterfalls / near: skull)

Discussion: which prompt dominates far vs. near, and why. In principle the **waterfalls** prompt should win *from a distance*, since it supplies the low spatial frequencies that set the overall shape and layout (our eyes weight low frequencies more when an image is small or blurry), while the **skull** prompt should win *up close* through the high frequencies — fine texture and edges we only resolve at short range. Applying the low-pass/high-pass split to the *noise estimate* rather than to the pixels builds this separation directly into the generated image. To test the far reading we downsampled our result to 16×16 : a faint bright vertical form does emerge against the dark surround and could pass for a fall, but the skull still dominates even at that size, so the hybrid effect here is real but only partial rather than the clean two-subject switch we were aiming for. The trick works best when the two subjects share a similar rough outline so the low-frequency prompt can set the shape and the high-frequency prompt just adds texture on top; a better result here would likely need different seeds or a low-frequency prompt whose silhouette matches the skull more closely.

Appendix: Prompts and Parameters Used

Task	Prompt(s)	Key parameters
0.1	old man / campfire / rocket ship	stage 1 & 2 num_inference_steps=20; comparison 5 vs 50 (rocket)
1.1	n/a	$t \in \{250, 500, 750\}$
1.2a	"a high quality photo" (null)	$t \in \{250, 500, 750\}$
1.2b	"a high quality photo"	strided_timesteps = range(990, -1, -30) (34 steps), i_start=10
1.2c	"a high quality photo"	i_start=10 (one-step vs iterative)
1.3	"a high quality photo"	5 samples no-CFG + 5 samples CFG, $\gamma = 7$
2.1	"a high quality photo" (null)	i_start $\in \{1, 3, 5, 7, 10, 20\}$; campanile + web + hand-drawn
2.2	"a high quality photo"	provided mask (campanile) + custom mask on web image
2.3	"a rocket ship"	i_start $\in \{1, 3, 5, 7, 10, 20\}$

Task	Prompt(s)	Key parameters
3.1	"an oil painting of an old man" / "an oil painting of people around a campfire"	$\gamma = 7$, i_start=0
3.2	"a lithograph of waterfalls" (low) / "a lithograph of a skull" (high)	Gaussian blur k=9, $\sigma = 2.0$; $\gamma = 7$, i_start=0

CFG guidance scale used throughout Parts 2–3: $\gamma = 7$.